

LUEGDE-PAENBRUCH, Germany

WMO: 104330

Lat: 51.8664N

Lon: 9.2708E

Elev: 850

StdP: 14.25

Time Zone: 1.00 (EUC)

Period: 01-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.9	17.2	8.2	8.6	14.6	12.3	10.6	19.5	24.5	43.4	21.8	42.9	5.0	10	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	82.3	66.2	78.5	64.9	75.0	63.2	67.9	77.6	66.4	75.4	64.6	72.5	7.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	94.7	70.8	63.3	89.8	69.2	61.8	85.2	67.8	32.7	77.3	31.4	75.5	30.1	72.2	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.9	16.6	14.6	7.8	87.2	7.3	4.9	2.6	90.7	-1.7	93.5	-5.8	96.2	-11.1	99.8		
			7.5	70.9	7.1	1.8	2.3	72.2	-1.8	73.2	-5.8	74.2	-11.0	75.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	32.8	33.2	39.2	47.8	53.3	59.1	62.8	62.3	56.2	48.6	41.4	33.5	
	DBStd	12.87	8.71	8.36	7.98	7.24	6.18	5.91	5.97	5.13	5.61	6.78	6.22	7.60	
	HDD50	2402	532	471	340	124	38	3	0	0	9	110	264	511	
	HDD65	6458	997	890	800	515	365	194	117	116	269	510	709	976	
	CDD50	1523	0	1	5	59	140	277	396	380	195	65	5	0	
	CDD65	104	0	0	0	0	2	17	48	31	6	0	0	0	
	CDH74	945	0	0	0	3	19	133	473	273	43	1	0	0	
	CDH80	210	0	0	0	0	0	17	117	74	2	0	0	0	
Wind	WSAvg	7.1	8.9	8.0	7.5	6.9	6.6	6.2	5.9	5.8	5.9	7.1	7.8	8.1	
Precipitation	PrecAvg	35.2	3.4	2.7	2.6	2.0	2.6	2.8	3.4	3.2	2.8	2.9	3.2	3.5	
	PrecMax	50.2	6.9	6.3	6.5	4.4	6.9	4.6	7.2	6.3	6.8	9.6	5.6	7.6	
	PrecMin	26.9	0.4	0.8	0.3	0.2	0.9	1.4	1.5	1.3	0.4	0.8	0.2	1.4	
	PrecStd	6.1	1.6	1.4	1.5	0.9	1.5	0.8	1.5	1.4	1.5	1.7	1.5	1.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.7	54.3	61.7	73.5	76.8	82.3	87.7	90.0	78.7	68.6	57.2	51.5	
		MCWB	49.4	47.2	50.7	56.9	64.4	67.3	66.3	66.2	64.7	61.2	52.9	48.9	
	2%	DB	49.7	50.0	57.6	69.0	72.1	78.0	83.4	81.4	73.9	63.6	54.0	47.6	
		MCWB	47.6	46.8	48.5	55.4	60.1	65.9	66.6	65.3	62.7	57.8	50.9	45.2	
	5%	DB	47.1	47.0	54.2	64.8	68.8	74.1	79.4	76.6	70.0	60.4	51.9	45.7	
		MCWB	45.0	44.3	47.7	53.2	57.3	63.4	64.9	63.9	60.8	56.2	49.4	43.8	
	10%	DB	44.4	44.4	50.6	60.8	65.4	70.4	75.3	72.5	66.2	57.9	49.7	43.4	
		MCWB	42.5	41.9	45.6	51.7	56.3	61.3	63.3	62.5	59.2	54.7	47.5	41.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.2	49.0	52.7	58.5	66.1	70.6	70.0	69.3	67.0	61.4	53.7	49.3
			MCDB	51.6	52.1	58.6	69.5	73.4	79.3	81.3	80.0	74.0	67.2	56.1	51.7
2%		WB	47.9	46.9	50.6	56.4	62.7	67.0	67.8	67.3	63.6	58.8	51.5	45.8	
		MCDB	49.6	49.6	55.4	66.1	69.4	75.8	78.3	76.7	71.7	62.2	53.6	47.0	
5%		WB	45.2	44.6	48.3	54.5	59.9	64.7	66.3	65.5	61.8	56.8	49.7	43.9	
		MCDB	46.7	46.8	53.1	63.1	66.3	71.7	76.0	73.8	68.7	59.9	51.8	45.7	
10%		WB	42.6	41.9	45.9	52.5	57.3	62.6	64.6	63.6	59.9	54.9	47.7	41.6	
		MCDB	44.4	44.2	50.4	59.7	63.0	68.5	72.9	70.6	65.3	57.5	49.6	43.1	
Mean Daily Temperature Range		5% DB	MDBR	7.1	8.2	11.8	15.6	15.7	16.0	16.6	15.4	14.6	11.2	7.4	6.9
			MCDBR	8.5	9.8	18.2	23.1	21.7	21.8	24.7	23.3	21.1	14.1	8.8	7.7
	MCWBR		7.9	7.4	11.3	12.2	11.9	11.1	10.0	10.6	11.2	9.3	6.7	6.8	
	5% WB	MCDBR	8.2	9.0	15.4	19.8	19.1	19.5	21.3	18.8	18.4	13.1	8.2	7.8	
		MCWBR	8.0	7.3	10.1	11.4	11.7	10.6	9.8	9.7	10.5	9.4	6.7	7.1	
Clear-Sky Solar Irradiance	taub	0.309	0.332	0.367	0.403	0.406	0.405	0.418	0.412	0.379	0.349	0.330	0.306		
	taud	2.391	2.364	2.288	2.229	2.251	2.273	2.243	2.278	2.364	2.431	2.408	2.378		
	Ebn at Noon	222	249	260	264	269	270	263	258	254	239	211	201		
	Edh at Noon	19	26	33	40	41	40	41	37	31	24	19	16		
All-Sky Solar Radiation	RadAvg	212	427	772	1250	1514	1637	1558	1322	956	555	249	161		
	RadStd	28	66	98	153	162	140	209	115	127	56	34	18		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (41 neighbors)	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	-229	N/A	N/A	

Nomenclature: See separate page